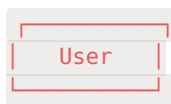


ER SCHEMA STRONG AND WEAK ENTITIES

STRONG ENTITIES

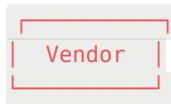
1. User

A person who created an account on your platform. They exist in the system even if they never made a single booking. Rahul Sharma signed up → he exists. He hasn't booked anything yet → he still exists. Fully independent.



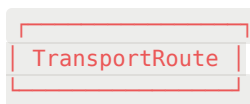
2. Vendor

IndiGo Airlines registered on your platform. They exist as a vendor regardless of whether they've listed any routes or hotels yet. Independent existence.



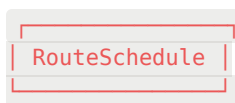
3. TransportRoute

The Mumbai → Delhi route operated by IndiGo exists as a concept in your system. It doesn't need a booking to exist. It doesn't even need a schedule yet. The route itself is an independent thing.



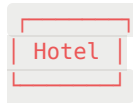
4. RouteSchedule

The 6AM timing slot on that Mumbai → Delhi route. It exists independently — it's a product that IndiGo has defined. No one needs to have booked it for it to exist.



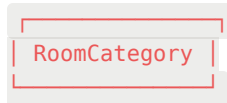
5. Hotel

The Taj Mahal Palace Mumbai exists in your system as soon as the vendor registers it. No booking needed for it to exist.



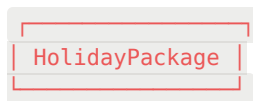
6. RoomCategory

The "Deluxe Sea View Suite" at Taj Mumbai. It's a product that the hotel has defined. Exists independently of any booking.



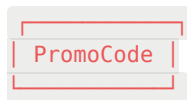
7. HolidayPackage

"Goa Beach Escape — 4 nights" exists as a product the vendor has listed. No one needs to have booked it.



8. PromoCode

"SUMMER20" exists in the system as soon as the admin creates it. It exists regardless of whether any user has used it yet.

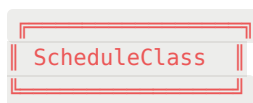


WEAK ENTITIES

9. ScheduleClass

Economy class with 180 seats at ₹3,500 base fare — this ONLY makes sense in the context of a specific RouteSchedule. If you delete the 6AM Mumbai → Delhi schedule, this class has nothing to belong to. It cannot exist alone.

Parent: `RouteSchedule`



10. ScheduleDeparture

"6E-201 on 15th March 2025, Economy, 43 seats left at ₹4,200" — this is a specific dated instance. Without the RouteSchedule and ScheduleClass it belongs to, this departure means nothing. It dies with its parent.

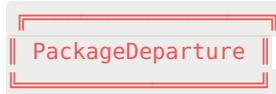
Parent: RouteSchedule + ScheduleClass



11. PackageDeparture

"Goa Beach Escape departing 1st March, 20 seats, ₹19,500" — completely meaningless without the HolidayPackage it belongs to. Delete the package, this evaporates.

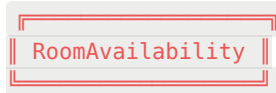
Parent: HolidayPackage



12. RoomAvailability

"Deluxe Suite on 20th March has 3 rooms left" — this is just a number attached to a room category on a date. No independent meaning whatsoever. Entirely dependent on RoomCategory.

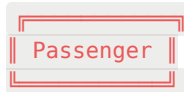
Parent: RoomCategory



13. Passenger

"Rahul Sharma, Age 28, Passport IN123456" listed on a transport booking. Without the TransportBooking record, this passenger record is an orphan — it means nothing. Who did he travel with? What flight? Unknown. Cannot exist alone.

Parent: TransportBooking (which you'll define in Step 4)



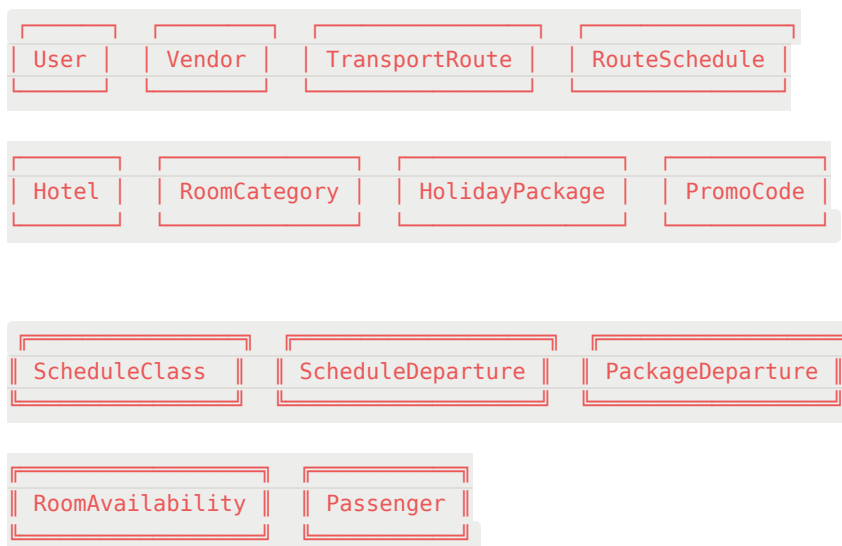
What About Booking, TransportBooking, HotelBooking, PackageBooking, Insurance, CancellationRefund, Review, DemandTag?

These are NOT entities you draw as rectangles in Step 1.

`Booking, TransportBooking, HotelBooking, PackageBooking` → these are **relationships** between entities. In ER modelling, the act of a User booking a ScheduleDeparture is a *relationship*, not a separate thing. They'll appear as **diamonds** in your diagram in Steps 4 and 5. Some of them will become tables in SQL (because they carry their own data), but in the ER diagram they are relationships first.

`Insurance, CancellationRefund, Review, DemandTag` → these are **derived/associative entities** that emerge from relationships. They come in Step 6 and 7 when you draw the features layer.

This is exactly the **aggregation concept** from your Image 2 — Booking is the aggregated relationship that then becomes an entity for Insurance and CancellationRefund to connect to.



- *"Strong — exists independently"*
- *"Weak — depends on [parent name]"*

#	Entity	Type	Why
1	User	Strong	Exists without any booking
2	Vendor	Strong	Exists without listing any service
3	TransportRoute	Strong	A route exists as a product independently
4	RouteSchedule	Strong	A schedule exists as a product independently
5	Hotel	Strong	A hotel exists without bookings
6	RoomCategory	Strong	A room type exists without bookings
7	HolidayPackage	Strong	A package exists without bookings
8	PromoCode	Strong	A code exists without being used
9	ScheduleClass	Weak	Dies without RouteSchedule
10	ScheduleDeparture	Weak	Dies without RouteSchedule + ScheduleClass
11	PackageDeparture	Weak	Dies without HolidayPackage
12	RoomAvailability	Weak	Dies without RoomCategory
13	Passenger	Weak	Dies without TransportBooking